



REVIEW

Optimizing the Ocular Surface Before, During, and After Glaucoma Surgery: A Narrative Review

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ABSTRACT

This narrative review synthesizes the published literature on ocular surface disease (OSD) and glaucoma filtration surgery to develop practical, evidence-informed clinical and surgical recommendations for optimizing ocular surface management before, during, and after surgery. The reviewed studies examine how OSD affects ocular inflammation and wound healing, how

chronic preservative-containing topical therapy affects bleb survival, and how reducing cumulative preservative load benefits ocular surface health. In patients with glaucoma, chronic use of preservative-containing topical medications can aggravate OSD, with approximately 50% of patients receiving topical intraocular pressure-lowering therapy reporting symptoms of OSD. OSD-related inflammation and tissue changes may adversely affect wound healing and compromise the success of glaucoma filtration surgery. OSD in patients likely to require filtration surgery should be actively identified and treated throughout the preoperative, intraoperative, and postoperative stages. Practical steps include screening for OSD, reducing exposure to preservative-containing medications, optimizing the ocular surface preoperatively, and minimizing conjunctival inflammation perioperatively. These strategies can be readily adopted into clinical practice and may help optimize conditions for bleb survival, although direct evidence demonstrating improved bleb survival is limited.

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Key Summary Points

Ocular surface disease (OSD) is highly prevalent in patients with glaucoma receiving long-term topical therapy and may adversely affect the success of glaucoma filtration surgery.

Chronic exposure to preservative-containing glaucoma medications promotes ocular surface inflammation and tissue changes that can impair wound healing and increase the risk of bleb failure.

This narrative review synthesizes current evidence and proposes five practical, evidence-informed recommendations for optimizing the ocular surface before, during, and after glaucoma filtration surgery.

Key strategies include identifying and treating OSD, minimizing preservative exposure, controlling perioperative inflammation, employing meticulous surgical techniques, and preserving long-term bleb function through postoperative ocular surface management.

Integrating these recommendations into routine clinical practice may help optimize conditions for bleb survival, support surgical outcomes, and facilitate long-term management of patients undergoing glaucoma filtration surgery.

INTRODUCTION

Ocular surface disease (OSD) is a spectrum of conditions that affect various tissues including the cornea, conjunctiva, eyelids, and lid margins, leading to dysfunction of the ocular structures responsible for tear film production. This dysfunction causes tear film hyperosmolarity and instability, leading to a self-perpetuating cycle of ocular surface inflammation and concurrent damage. The use of chronic preserved topical ophthalmic formulations is a significant risk factor for developing OSD that escalates with increasing number of medications

[1]. In chronic conditions such as glaucoma, which require long-term topical therapy, the prevalence of OSD increases with age, severity of disease, number of eyedrops, and the duration of topical therapy [1, 2]. As intraocular pressure (IOP) is the only modifiable risk factor in the management of glaucoma, patients with moderate-to-severe glaucoma often have an increased burden of medications due to multidrop therapy, increasing the risk of OSD [2]. For example, 38% of patients treated with one drop/day of preserved antiglaucoma medication had moderate-to-severe OSD, which increased to 54% and 71% when treated with 2 drops and ≥ 3 drops, respectively [2].

In recent years, there has been increasing emphasis on minimizing cumulative exposure of the ocular surface to preservatives and active compounds throughout the natural history of glaucoma. The LIGHT trial and subsequent studies have established selective laser trabeculoplasty (SLT) as an effective first-line treatment capable of providing durable IOP control while delaying or avoiding the need for chronic topical therapy [3, 4]. Beyond its role as a primary intervention, SLT can also serve as a treatment-sparing strategy, allowing reduction in the number of topical medications in patients already receiving multiple agents or in those experiencing OSD related to chronic drop use [3, 5]. In the post-LIGHT era, early adoption of SLT represents an important strategy for preserving ocular surface integrity and reducing the risk of drop-induced OSD, with potential benefits for future surgical outcomes. [6]

Although preventing vision loss and maintaining quality of life are the primary goals of glaucoma management, concomitant OSD has a significant deleterious effect on quality of life and reduces adherence, resulting in faster disease progression [7], and higher chances of surgical intervention [8]. OSD is therefore a major barrier to effective glaucoma management, compromising long-term outcomes.

Glaucoma filtration surgery is generally reserved for patients with glaucoma that is inadequately controlled despite maximal tolerated medical and/or laser therapy, including those with advanced disease in whom a sufficiently low target IOP cannot be achieved medically,

in accordance with European Glaucoma Society guidelines [9]. Such patients have typically been using multiple antiglaucoma drops for a long duration of time, which may compromise outcomes of subconjunctival glaucoma surgery. Toxicity due to chronic, long-term topical glaucoma therapy results in alterations to the ocular surface, including inflammation, epithelial damage and tear film instability changes which maybe subclinical [10, 11]. These alterations can precipitate more aggressive wound healing responses following surgery, obstructing aqueous humor (AH) outflow [10]. Previous studies have shown patients with OSD have poorer outcomes after glaucoma filtering surgery [1, 12]. Additionally, a higher preoperative daily dose of the preservative benzalkonium chloride (BAK) (e.g., more daily topical medications) resulted in a shorter time to surgical failure in patients who underwent trabeculectomy [13]. Furthermore, if topical antiglaucoma medications need to be reintroduced following surgery, for example in a partially functioning bleb, there is a risk that further drop-induced inflammation may convert a partially successful operation to a failed filter. However preoperative topical steroid use can increase success rates [14, 15].

Although there is abundant literature highlighting the importance of managing OSD prior to glaucoma surgery, there is little examining the ocular surface management throughout the peri-operative stages. In this narrative review we present five tips to optimize the success of filtration surgery for glaucoma by managing OSD preoperatively, intraoperatively, and postoperatively [11, 16, 17]. This article is intended to provide practical, evidence-informed guidance recommendations for clinicians. Relevant publications were identified through targeted searches of the published literature and selected based on their relevance to the topics discussed. As a narrative review, no formal systematic search strategy or quality assessment was undertaken.

Before outlining these recommendations, it is useful to review key principles of subconjunctival glaucoma surgery.

Glaucoma Surgery Outcome: The Filtering Bleb

Subconjunctival glaucoma surgery relies on enhancing drainage of AH by creating a fistula from the anterior chamber into the subconjunctival/sub-Tenon's space that creates an area of elevated conjunctiva called a bleb. A functioning bleb is fundamental to the success of traditional filtering surgery such as trabeculectomy and glaucoma drainage device implantation and newer techniques such as new bleb forming procedures. The success of filtering surgery relies on controlling the postoperative wound healing response. Inadequate control of pre-existing ocular surface inflammation or surgically induced inflammation can lead to clot formation, angiogenesis and fibrosis resulting in bleb failure.

Studies using *in vivo* optical coherence tomography (OCT) imaging of the anterior segment indicate that successful blebs are diffuse and present various structural characteristics such as an inhomogeneous architecture with a thick wall, a suprascleral fluid space of low reflectivity, as well as an adequate bleb height [18]. Ocular tissue fibrosis at the bleb site may decrease egress of AH or cause a complete blockage of outflow, resulting in poor IOP control leading to surgical failure. Insights using imaging have shown that a functioning bleb involves a compromise between physiologic wound healing and the scleral outflow of AH to maintain target IOP [19–21].

Studies using *in vivo* confocal microscopy indicate that normal functioning bleb anatomy involves various structural changes including the formation of numerous intraepithelial microcysts, widely spaced subepithelial connective tissue and minimal vascularity [19–21]. These microcysts are formed by transconjunctival outflow of AH. It has even been suggested that these microcysts may correspond to goblet cells that contain AH [22]. In addition, high preoperative conjunctival goblet cell density has also been correlated to successful filtering surgery and good IOP control [23, 24]. However, BAK decreases conjunctival goblet cells survival and consequently, patients treated with

BAK-containing eyedrops have a reduced preoperative goblet cell density [25]. Hence, pre-existing inflammation from preservative-containing multidrop glaucoma therapy is a risk factor for surgical failure.

Advances in imaging are providing novel methods of evaluating the ocular surface and associated structures before and after glaucoma surgery. For example, in vivo confocal microscopy allows quantification of goblet cell density in the conjunctiva. Optical coherence tomography-angiography also provides perioperative conjunctival vascularity [26], while conventional anterior segment optical coherence tomography can be used to provide quantitative measures of bleb morphology, including bleb height, fluid cavity height, bleb wall thickness and microcyst density. These metrics can be assessed over the postoperative period to determine changes in bleb architecture that may indicate bleb failure and warrant intervention [18].

Emerging evidence highlights the pivotal role of the conjunctival lymphatic system in maintaining filtration bleb function by providing an auxiliary outflow route for AH [27]. Experimental and clinical imaging studies demonstrate organized lymphatic channels around functioning blebs, whereas reduced or disorganized lymphatic patterns are associated with surgical failure [20, 28]. Early postoperative lymphangiogenesis may aid aqueous absorption and pressure control, while excessive fibrosis can compromise bleb drainage [29, 30]. Advances in anterior segment OCT and in vivo confocal microscopy now permit visualization and quantification of conjunctival lymphatic vessels, which may offer potential biomarkers and therapeutic targets for optimizing bleb survival [31].

Protecting and Preparing the Ocular Surface Before Glaucoma Surgery

TIP 1: Look for Ocular Surface Disease Preoperatively

Assessment for ocular surface disease should be a key element of the glaucoma examination at all stages of disease management. This

is particularly important before considering glaucoma surgery and is likely to influence the choice of procedure, timing and preparation required prior to surgical intervention. OSD is highly prevalent in individuals with glaucoma, with approximately 50% of patients with glaucoma using topical therapy reporting symptoms of OSD [1]. The prevalence is likely to be higher in the surgical population as subconjunctival surgery tends to be reserved for cases where topical medications have failed, and many of those requiring surgery have been using multiple eye drops for a long duration. Surgery may also be considered as an alternative to eye drops in patients with iatrogenic drop-induced or drop-exacerbated OSD [25]. It is imperative that the evaluation of OSD includes assessment of symptoms and signs, as there may be a disconnect between perceived symptoms and clinical appearance. Recommendations from DEWS III propose screening for OSD using a standardized patient questionnaire followed by clinical evaluation, including assessment of non-invasive breakup time, osmolarity and ocular surface staining with fluorescein [32, 33]. Questionnaires include the ocular surface disease index (OSDI), however the OSDI lacks specificity in detecting OSD in patients with glaucoma due to its inclusion of questions related to visual function, which may be affected by glaucoma rather than OSD [34]. The more specific FAST® questionnaire, specially designed for patients with glaucoma, can also be used in clinical practice to reliably detect patients at risk of OSD [35]. Signs such as conjunctival hyperemia and eyelid redness are suggestive of ocular inflammation [32, 33, 35]. Corneal and conjunctival staining are beneficial for determining disease severity in severe dry eye disease (DED) that is often present in patients undergoing glaucoma filtration surgery [32].

TIP 2: Optimize the Ocular Surface Prior to Surgery

Steps should be taken to remove iatrogenic causes of ocular surface inflammation, which may compromise surgical success. Measures include switching to unpreserved topical medications and decreasing the number of eyedrops.

This is especially important if the patient is using topical medications that are known to be harmful to the ocular surface such as brimonidine [36, 37]. Although preservatives are considered the main contributors to OSD through cumulative dose-dependent toxicity, active ingredients such as brimonidine may induce distinct ocular surface changes, including delayed hypersensitivity reactions, which require discontinuation of the causative agent rather than simply reducing preservative exposure [36]. Where feasible, surgeons can elect to halt all topical therapy and switch to oral acetazolamide before surgery and special consideration should also be given to minimizing preservative exposure from time of diagnosis in those likely to need glaucoma surgery.

Chronic ocular surface inflammation is thought to promote subconjunctival fibroblast activation and profibrotic signaling, thereby increasing the risk of postoperative bleb fibrosis and surgical failure [29].

BAK is the most common preservative in eye drops, however, it causes damage to the ocular surface and the trabecular meshwork through dose- and time-dependent toxicity [23]. Higher exposure to BAK-containing ophthalmic formulations has been associated with earlier failure of trabeculectomy and decreased success of glaucoma surgery [13]. Halting the use of preservative-containing medications can improve the ocular surface in about 2 weeks due to the decreased inflammation and, perhaps, increased goblet cell density [25, 38]. Halting all preserved medications may not be feasible; hence, minimizing the load of preserved medications is still beneficial as it decreases the cumulative load of preservative that is cytotoxic to the cornea and conjunctiva. For example, the threshold concentration of BAK at which cytotoxicity occurs is 0.005%, and topical eyedrops typically contain BAK concentrations of 0.01–0.02% [39].

OSD such as DED and blepharitis are extremely common among patients with glaucoma, and efforts should be made to treat these conditions pre-operatively. Ideally ocular surface inflammation should be controlled at least 1 month prior to surgery [12]. The pre-operative use of preservative-free topical lubricants is beneficial for addressing dry eye thus decreasing pre-existing inflammation [11]. Given the

role of inflammation in surgical failure, the use of preservative-free topical lubricants, concurrently with topical glaucoma medications, is recommended [40]. Notably, topical lubricants containing BAK decrease goblet cell viability, which is a factor of filtering surgery failure [38, 40, 41].

Inflammation can be controlled with topical fluorometholone (FML) or low-dose hydrocortisone, as these agents confer a lower likelihood of inducing an IOP spike or cataract compared to more potent topical corticosteroids [14, 15, 42, 43]. In selected patients with moderate-to-severe ocular surface disease undergoing bleb-forming glaucoma surgery, a short preoperative course of low-potency, preferably preservative-free topical corticosteroids may be considered [44]. Preoperative steroids reduce conjunctival inflammatory cell density and may decrease the need for postoperative bleb interventions, although evidence demonstrating improved long-term surgical success remains limited [14, 15, 31].

Alternately, or adjunctively, topical non-steroidal anti-inflammatory drugs (NSAIDs) or cyclosporine can be initiated to suppress ocular surface inflammation [11, 25]. Where feasible, preservative-free ophthalmic formulations should be prioritized. Treatment selection should also consider patient preference, ease of administration, local availability, and access to therapy to support long-term adherence.

Blepharitis contributes to tear-film instability, ocular surface inflammation, and increased risk of bleb-related infection after trabeculectomy. Patients, particularly those with moderate-to-severe disease, should undergo preoperative management with daily warm compresses and eyelid hygiene. When indicated, antibiotic therapy may be added, with topical antibiotics (e.g., erythromycin, bacitracin ointment, or azithromycin) recommended for anterior blepharitis, while oral tetracyclines or macrolides may be considered because of their additional anti-inflammatory effects [45].

In conclusion, preoperative optimization of the ocular surface through reduced preservative exposure and control of inflammation promotes wound healing, decreases complications, and improves long-term surgical success.

Management of the Ocular Surface During Glaucoma Surgery

Trauma due to surgery results in inflammation that is mediated by a cascade of events involving clotting and the release of hormones (e.g., prostaglandins), cytokines and growth factors (e.g., fibroblast growth factor), which attracts components of the immune system (e.g., neutrophils, macrophages) to the surgical site [46, 47]. Subsequent migration of fibroblasts to the surgical site during tissue repair and remodeling, results in fibrosis and scar formation [46].

Treating dry eye and blepharitis before surgery ensures a healthier conjunctiva and improves the likelihood of long-term surgical success.

TIP 3: Respect the Conjunctiva During Surgery

Intraoperatively the surgeon can mitigate inflammation by controlling bleeding, meticulous technique, gentle handling of tissue and limiting interventions [48]. Minimally invasive bleb surgery (MIBS), including PreserFlo® or Xen®, may be good alternatives as these procedure are less invasive than traditional glaucoma surgery and have some advantages over traditional filtration surgery, including less tissue manipulation and shorter surgery times [36]. However, less invasive techniques tend to create a smaller fistula and do not allow the same room for bleb manipulation and postoperative titration of IOP, for example, through the use of releasable sutures or laser suture-lysis. Studies are required to determine whether these features of MIBS [49], may decrease inflammation and the risk of postoperative fibrosis.

Gentle surgical technique with minimal conjunctival trauma, and the use of MIBS when feasible, help reduce postoperative inflammation, preservative conjunctival integrity, and promote faster recovery.

TIP 4: Control Cell Proliferation to Prevent Fibrosis

Antimitotic agents such as 5-fluorouracil (5-FU) and mitomycin C (MMC) are primarily used

intra- and postoperatively during glaucoma filtration surgery to inhibit fibroblast activity and reduce the risk of postoperative fibrosis. In selected high-risk cases, preoperative modification of topical therapy with short-term topical corticosteroids has been shown to reverse adverse conjunctival changes induced by chronic antiglaucoma medications and may improve surgical outcomes [31, 50]. Although 5-FU has been routinely used during filtration surgery, this antimitotic is toxic to the cornea and conjunctival epithelium in high doses which can cause vision threatening complications [47, 51]. More commonly, MMC is now used during glaucoma surgery for preventing fibrosis. Comparisons of MMC to 5-FU in glaucoma surgery indicate less epitheliopathy and lower risk of postoperative corneal damage with MMC [52]. Higher concentrations of MMC and application over a larger area of the surgical site have been associated with reduced postoperative fibrosis after filtration surgery, although the magnitude of benefit appears to depend on baseline scarring risk, concentration range, and surgical context [53]. In contemporary practice, lower MMC concentrations (e.g., 0.02%) may be insufficient in eyes at increased risk of scarring, while higher concentrations (e.g., 0.04%) are more commonly employed in such cases and have been associated with improved surgical success. Such dosing may therefore be appropriate for selected high-risk patients, including those with a preoperative diagnosis of ocular surface disease undergoing filtration surgery. However, increasing concentration or extending contact time also increases the cytotoxic effects of MMC and carries the risks of corneal decompensation, blebitis and endophthalmitis [46, 51].

In addition to its effects on the corneal and conjunctival epithelium, MMC may also damage the meibomian glands. It is well known to be toxic to ocular surface tissues, with potential adverse effects on adjacent eyelid structures [16, 54]. These alterations can exacerbate evaporative dry eye and destabilize the tear film, worsening the ocular surface postoperatively, especially in patients with pre-existing OSD [16]. Therefore, minimizing MMC exposure and using the lowest effective dose and contact time is advisable

in those with compromised meibomian gland function or dry eye disease.

Intraoperative injection of MMC addresses some of these issues and has some advantages over soaked sponges, including delivery of a standardized dose to the surgical site, no risk of retained sponge fragments and less surgical time [55]. Notably, the dose of MMC that is delivered to the ocular tissues remains unknown with soaked sponges. Additionally, the use of soaked sponges prolongs surgery time as they are applied intraoperatively up to 5 minutes [56]. Some have postulated that perioperative MMC injection may be more appropriate in patients with pre-existing OSD [57]. Perioperative MMC injection (0.002–0.02%) has become more common over time for trabeculectomy and glaucoma drainage devices [46, 49, 58]. Perioperative MMC injection is also reported for implantation of some newer bleb forming devices [49]. More recently, the use of anti-vascular endothelial growth factor (anti-VEGF) during filtration surgery has also been investigated. Less bleb vascularity has been reported with subconjunctival delivery of anti-VEGF agents compared to MMC, and adjunctive use of anti-VEGF and MMC may have complementary effects on modulating wound healing [59]. A prospective randomized double-blinded clinical trial has shown that intracameral bevacizumab improves success rates and reduces needling rates after trabeculectomy [60].

Preserve the Bleb

TIP 5: Preserve the Bleb and Ocular Surface

Long-term preservation of the bleb and optimal outcomes after glaucoma filtration surgery require lifelong follow-up to detect and manage complications early. Evidence indicates a dynamic interaction between the ocular surface and filtration bleb, where postoperative inflammation and ocular surface disease can accelerate fibrosis and bleb failure [16]. Routine ocular surface evaluation and treatment with preservative-free lubricants, anti-inflammatory, or immunomodulatory agents can help stabilize the tear film, reduce inflammation, and maintain bleb

function [61, 62]. Minimizing toxic exposures and managing meibomian gland dysfunction (MGD) are equally important. Careful, individualized use of antimetabolites such as MMC or 5-FU, at the lowest effective dose and with delivery tailored to patient risk, can further prevent fibrosis, and support long-term surgical success.

Postoperative follow-up should include assessment of both bleb morphology and ocular surface health. Intensive postoperative corticosteroid use limits the inflammatory response and reduces fibrosis of the bleb. Interventions such as scleral flap suture adjustment, laser suturelysis, bleb massage and needling can be used to titrate flow and counter fibrosis. Longer term preservative-free steroid therapy can be prescribed to further limit late-stage bleb fibrosis. Postoperatively, subconjunctival MMC or 5-FU injections may also be used to further limit the wound healing response.

If the desired IOP target is not achieved and topical medications must be reintroduced, preservative-free preparations are highly recommended to minimize the risk of inciting further inflammation postoperatively.

Postoperative evaluation of the eyelids and the ocular surface is essential, as eyelid margin disease, particularly blepharitis and MGD, can exacerbate inflammation, destabilize the tear film, and compromise bleb function [63]. Routine management of these conditions should be part of postoperative care [16]. Measures such as lid hygiene, warm compresses, and targeted MGD therapy help maintain ocular surface stability and patient comfort [64].

Preservative-free formulations are an important adjunct in postoperative care following ocular surgery, helping to maintain ocular surface integrity, reduce friction with the filtering bleb, and alleviate discomfort [65]. They are particularly beneficial in eyes with pre-existing ocular surface disease from long-term antiglaucoma therapy [66]. The choice of lubricant should be guided by the predominant ocular surface abnormality [67, 68]. Preservative-free formulations are essential to avoid epithelial toxicity and inflammation that may compromise bleb function.

Long-term success of filtration surgery relies on vigilant postoperative care and preservation

of ocular surface health through consistent, preservative-free management of inflammation and tear film stability.

DISCUSSION

This article presents a structured, evidence-informed framework for managing ocular surface disease across the perioperative stages of glaucoma surgery. In contrast to prior reviews that have focused primarily on preoperative optimization, it consolidates recent findings on imaging biomarkers, lymphatic modulation, and preservative minimization within a unified approach to support long-term surgical success.

Controlling inflammation both pre- and postoperatively is essential for reducing fibrosis and encouraging long-term bleb survival. Inflammation of the ocular surface is, however, often present pre-operatively and is associated with OSD. We propose guidance for pre-operative, intra-operative and postoperative measures to reduce the risk of failure of filtration surgery due to OSD (Fig. 1). Managing OSD concurrently with each phase of glaucoma management is fundamental for long-term success.

Most of the evidence supporting ocular surface optimization derives from anterior

bleb-forming procedures, such as trabeculectomy, and extrapolation of these strategies to posterior bleb-forming surgeries, including tube shunts, should be interpreted with caution given differences in surgical anatomy and wound-healing dynamics. In addition, evidence remains limited for minimally invasive bleb-forming procedures and diverse patient populations, and further studies are needed to establish the generalizability of these recommendations.

One issue that remains incompletely addressed is the optimal timing of surgical intervention, which often depends on surgeon experience and patient-specific factors. Though surgery exposes the patient to a risk of complications, there is also a risk associated with conservative or medical treatment, including the risk of glaucoma progression. In the absence of formal recommendations, timing should be individualized based on IOP control, rate of disease progression, and ocular surface condition, as earlier intervention with selected procedures has been shown in some studies to improve IOP, ocular surface health, and quality of life compared with delayed intervention [69]. Further studies are needed to determine whether the safety and efficacy profiles of these techniques support earlier intervention before the onset of significant inflammation or OSD.

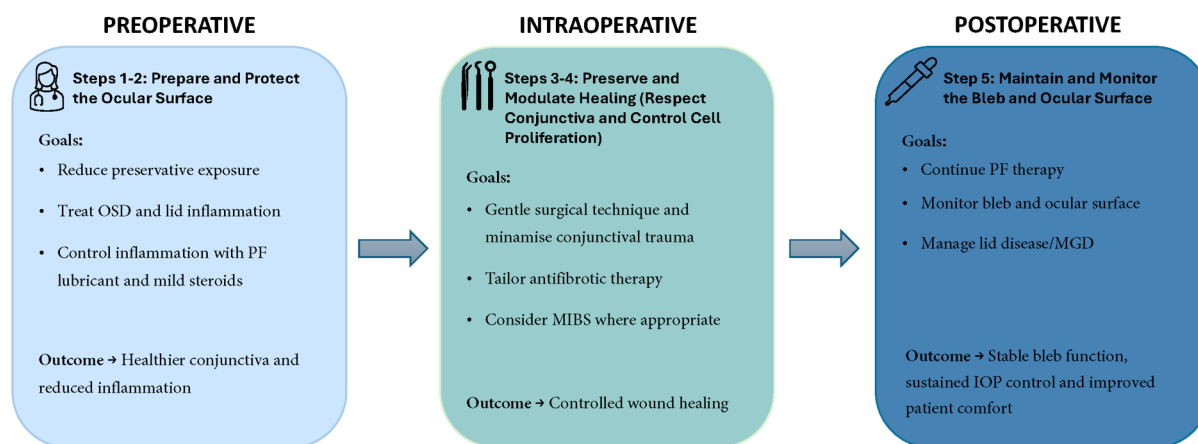


Fig. 1 Phase-based approach to ocular surface optimization in glaucoma surgery, highlighting preoperative preservation, intraoperative protection, and postoperative maintenance to promote wound healing, bleb function, and

long-term surgical success. *IOP* intraocular pressure, *MGD* meibomian glands dysfunction, *MIBS* minimally invasive bleb surgery, *OSD* ocular surface disease, *PF* preservative-free

It should also be acknowledged that preoperative ocular surface optimization is unlikely to completely reverse the chronic conjunctival and Tenon's capsule changes induced by long-term exposure to preserved topical therapy [11]. While reducing active inflammation before surgery may improve the perioperative ocular surface environment and facilitate postoperative bleb management, these observations further support minimizing preservative exposure throughout the course of glaucoma management rather than relying solely on preoperative intervention.

CONCLUSIONS

Multiple strategies can improve the likelihood of achieving a long-term functioning bleb, including reducing pre- and postoperative exposure to preservatives in glaucoma medications and controlling inflammation and cell proliferation to limit fibrosis. Prospective studies evaluating the impact of interventions for OSD management on surgical outcomes remain limited. While individual primary studies have explored specific interventions, such as topical steroid use prior to glaucoma filtration surgery, heterogeneity in medication exposure, surgical techniques, outcome measures, and preoperative protocols precludes definitive conclusions [4]. Consequently, much of the available evidence is derived from reviews rather than primary studies. Therefore, the recommendations presented in this review should be considered evidence-informed and intended to support clinical decision-making in areas where high-quality prospective evidence remains limited. There is a clear need for consensus-based recommendations to standardize the identification and management of OSD before, during and after glaucoma surgery. This framework underscores the importance of ocular surface preservation as a continuous process before, during, and after surgery to maximize bleb survival and long-term vision outcomes.

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Declarations

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Ethical Approval. This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors.

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